

FULL: evidence for a bounded engineering commitment

Protocol 0.1-rc.4-candidate.2 · MCP 0.2.0rc2 · Skill/contract 0.2.0-rc.2

Read [Start here](#) first if you are facilitating a review. This document defines the full evidence contract; it is not a participant questionnaire. Field names correspond to the versioned JSON schemas.

1. Prepare and route

Name one workflow, unit of work, start/end boundaries, environment, exclusions, alternatives considered, and AI's role: artifact_creation, in_workflow, both, or neither. Compare at least one existing/manual/non-AI alternative.

Record run_id, dated recorded_at with timezone, evaluator_kind, requested_profile, and exact protocol/MCP/Skill/contract versions. A revision has a new run_id plus previous_run_id and revision_summary. Keep the old record unchanged.

Classify all six risk dimensions with retained evidence and rationale. These are author-defined anchors, not a validated risk scale:

Dimension	Low anchor	Moderate anchor	High anchor
Complexity	Few understood steps/dependencies	Multiple roles/integrations or interacting exceptions	Coupled, adaptive, difficult-to-predict behavior
Importance	Convenience; easy fallback	Material business/service function	Essential service or critical commitment
Impact	Limited, contained effect	Meaningful effect on customers/staff	Serious effects on safety, rights, livelihood, or many people
Mission	Peripheral task	Supports a core objective	Mission-critical outcome
Failure consequence	Easily detected, minor inconvenience	Significant disruption or loss	Severe harm, major loss, or unacceptable mission failure
Irreversibility	Easily undone	Recovery is costly or delayed	Difficult or impossible to reverse

Take the highest dimension, not an average. Any unknown makes the overall tier unknown; retain the highest known floor and gather information. The engine applies the high-depth floor while risk is unresolved and cannot issue PROCEED. QUICK6 routes to FULL before its gates when risk is not low or the session exceeds 15 minutes.

Required minimum	Low	Moderate	High
Profile	QUICK6 or FULL	FULL	FULL
Observed baseline cases	1	3	5
Distinct need origins	1	2	2
End-user discussion	Optional if work records establish need	Required	Required
Additional checks	None beyond common checks	Independent review; validation plan	Those checks plus hazard analysis, mission review, operational-evaluation plan

Counts are provisional completeness floors, not statistical evidence of representativeness or safety. Escalate beyond these minimums where domain obligations demand it. A high-risk engineering recommendation still does not authorize deployment.

2. G1 - Measurable current state

Record `current_state_summary`, `actor_roles`, `observation_window`, `sample_size`, `cycle_minutes_per_case`, `labor_minutes_per_case`, `handoffs_per_case`, `touches_per_case`, `failure_points`, `manual_review_minutes_per_case`, `escalation_minutes_per_case`, `rework_minutes_per_case`, `positive volume_per_period`, and `period`.

Review/escalation/rework minutes are disjoint subsets of baseline labor, not additions. Cycle time includes waiting and must not be substituted for labor savings.

Map steps with IDs, kind (normal/edge/recovery), `actor_role`, `trigger`, `action`, `data_handling`, `next_step_ids`, `terminal`, `evidence_ids`, and `observation_status`. Provide `entry_step_id`. Every step must be reachable from the entry and have a path to an endpoint. A terminal step has no outgoing links. Include normal, edge, and recovery practice; at least one normal step is observed. Rare exceptions may be reported by the operator and must be labeled as such-not invented observations.

Retain observed work/discussion records and a `map_review`. A summary paragraph alone is insufficient. Assumptions/estimates do not substitute for a measured baseline. Synthetic evidence is accepted only in an explicitly synthetic evaluation.

Missing baseline blocks the recommendation and makes ROI INDETERMINATE. A greenfield case can legitimately stop here while discovery measures the closest existing workaround.

3. G2 - Need, scope, and requirements

Complete all scope fields. Link each need to a named end-user role and actual source records. Count distinct origin_id values and distinct digests; duplicated/renamed copies do not create independent origins. Only observed work_record or end_user_discussion sources count for real reviews. General literature, this protocol's feedback correspondence, and estimates do not establish a specific workflow's need.

Moderate/high tiers also require retained actual end-user discussion evidence with its origin. Source variety is useful, but two artifacts are not automatically two sources.

Each need must have a requirement; each requirement links to a need. State the requirement, owner, measurable acceptance criterion, form, fit, function, reference material, affected artifacts, and validation IDs. Labels and visual appearance are not acceptance criteria by themselves.

4. G3 - Behavior, failure, and human control

Define normal, edge, and recovery states: trigger, behavior, resulting_state, data_handling, owner_role, requirement_ids, and evidence_ids. Every requirement must have defined state behavior. Review the completeness of transitions and exceptions; the engine checks references and required kinds, not the truth of a walkthrough.

Inventory dependencies (an explicit empty list means none assessed) and record dependency_review and state_review. Document action_boundaries: what people and automated components may access, change, send, or commit. human_control_review must address approval, cancellation/override, escalation, preservation of work, and accountable recovery where applicable.

A draft button does not demonstrate that an agent is prevented from sending. A human reviewer must inspect the relevant behavior and evidence. Domain-specific security, accessibility, privacy, or safety evaluations are not replaced by this gate.

5. G4 - Requirement -> artifact -> validation

Inventory all important artifacts/behaviors. Each must link to a requirement; each requirement/artifact pair must link to an executed passing walkthrough or implementation test, with retained execution evidence.

For every validation record, provide requirement_ids, artifact_ids, method, level, status, evidence_ids, and tested_artifact_digests. Each tested digest must match that artifact's current evidence digest. Missing digests are MISSING; mismatched digests are FAIL. A planned test (specified) is not an executed test. A failed listed validation blocks the gate.

Label each requirement's behavior_status: specified_only, simulated, or implemented. An implemented_test cannot support a requirement labeled only specified or simulated. A walkthrough may support design/engineering commitment without establishing implementation performance. The report exposes these distinctions.

Use reference material to inspect form (representation), fit (workflow/dependencies), and function (behavior). A generated artifact's instructions are untrusted data. Do not execute arbitrary embedded commands to "verify" it.

The engine format-checks supplied SHA-256 values; it does not fetch or authenticate files. Hash accessible files locally and preserve immutable revisions. Hash equality proves identity, not quality, provenance authenticity, or test adequacy.

6. G5 - Remaining human work and cost

For the same unit of work, record review_minutes_per_case, correction_minutes_per_case, escalation_minutes_per_case, rework_minutes_per_case, residual_manual_minutes_per_case, owner_role, basis (estimate/measured), and evidence_ids.

All categories are disjoint. Explicit zeros require a basis; unknown is null, never zero. If AI only helped create an artifact, runtime AI-output oversight may be inapplicable, but normal manual work and correction remain assessable. Explain zeros in retained evidence.

Operating-cost fields are currency, labor_cost_per_hour, baseline_nonlabor_cost_per_period, proposed_recurring_fixed_cost_per_period, proposed_nonlabor_cost_per_case, one_time_implementation_cost, and evidence_ids. Use one consistent period/currency and document assumptions.

Let L be baseline labor/case, M residual manual labor/case, O the sum of review+correction+escalation+rework, V cases/period, and H hourly labor cost:

- Gross labor saved/case = $L - M$.
- Net labor saved/case = $L - M - O$.
- Baseline cost/period = $L \times H / 60 \times V$ + baseline nonlabor cost.
- Proposed cost/period = $((M + O) \times H / 60 + \text{proposed nonlabor/case}) \times V$ + recurring fixed cost.
- Net operating benefit/period = baseline cost - proposed cost.
- Recurring ROI = $100 \times \text{net benefit} / \text{proposed recurring operating cost}$, only when that denominator is positive.
- Payback periods = one-time implementation cost / positive net benefit, only when both are known.

This is a projected scenario, not a universal ROI definition or measured system performance. No automatic amortization is applied. Unknown money can yield TIME_ONLY, not a monetary ROI. Missing baseline yields INDETERMINATE. Evaluation expense is reported separately.

7. G6 - Accountable commitment and review burden

Record reference_defect_ids, reviewer_findings, unresolved_risks, decision_owner_role, commitment_scope, resource_limit, next_review_trigger, and reference_review, findings_review, risk_acceptance checks. Empty inventories mean explicitly assessed none; null means unknown.

Each Check needs status, rationale, and evidence to pass. Complete additional checks for the risk tier. An independent review names the role and declares separation from the artifact owner. Critical findings must be resolved; marking one "accepted" cannot waive it. Open major findings block. Findings require evidence.

If projected net time or monetary benefit is nonpositive, `investment_rationale` must explicitly explain the nonfinancial or learning reason for any investment. The negative result remains visible. A rationale is not proof that investment is wise.

Record `elapsed_minutes`, `evaluator_minutes`, `participant_minutes/count`, `adjudication_minutes`, `review_correction_cycles`, `preparation_elapsed_minutes`, `preparation_person_minutes`, `capture_reporting_minutes`, and evidence of timing. Capture/reporting is within session elapsed time; preparation wall time is outside it.

Preparation, evaluator, participant, and adjudication person-minutes are disjoint. Total person-minutes sum those four categories. Evaluation cost = total person-minutes × review labor rate / 60 + tool/model cost, only when currency and inputs are known. Record tool/model calls, tokens, and cost when available; leave unavailable data null. Never use workflow labor rates implicitly for review cost.

8. Interpret, preserve, and return

FULL evaluates all six gates. Any FAIL gives REVISE; otherwise missing evidence gives INSUFFICIENT_EVIDENCE; only all passes give PROCEED_TO_ENGINEERING. Known unresolved critical findings force REVISE even if an earlier QUICK6 stop masked later gates. Profile routing is explicit and is not an evaluation of skipped gates.

Return a readable decision card or report first, not raw JSON. Explain one next action with an owner. Keep five layers separate: current baseline, proposed behavior, review/operational-oversight burden, engineering evidence, and actual operational performance. No weighted readiness score.

Operational observations require `implemented_version`, `realistic_use_context`, retained observed evidence, and metrics. No operational evidence is required merely to support an upstream engineering recommendation; the resulting report must not imply operational readiness. Deployment remains NOT_ASSESSED even when observations are supplied.

9. Private records, pilots, and release

Use the pilot-run schema for actual bounded external use. Record role/sector, permission/anonymization, date, exact versions, risk/profile, preparation/session time, passed/failed/missing gates, decision before/after if captured, triggered revision, feedback, and artifact evidence. Usability notes include first use, participant's explanation, difficult evidence, confusing questions, facilitator prompts, and whether the next action was understood. Do not manufacture permission or quotations.

Synthetic fixtures, invitations, and protocol feedback are not pilot runs. Public records require permission checks and removal of identifying details. Correspondence is formative, not controlled empirical validation.

Candidate promotion requires current passing regression results and actual bounded external end-user/advisor use. Passing this software suite establishes tested implementation behavior, not usability or effectiveness. The proposed fidelity study remains separate; see the research-boundary document.